

Direct variation



- 1 If $(4, 512)$ is on the curve, $P = kQ^4$, solve for k .
- 2 A variable M is directly proportional to the square of N .
 - a Using k as the constant of proportionality, form an equation for M in terms of N .
 - b Given that $(4, 64)$ satisfies the equation, solve for k .
- 3 Consider the relationship where y is directly proportional to the cube of x .
 - a Using k as a constant of proportionality, state the equation relating x and y .
 - b The following table of values shows the relationship between x and y :

x	1	2	3	4	5
y	3	24	81	192	375

Solve for k , the constant of proportionality.

- 4 y is a power function which is directly proportional to x^n .
 - a Using k as the proportionality constant, form an equation for y in terms of x .
 - b The table below shows values that satisfy the relationship $y = kx^n$, for the case when $n = 2$.

x	2	4	6	8	10
y	32	128	288	512	800

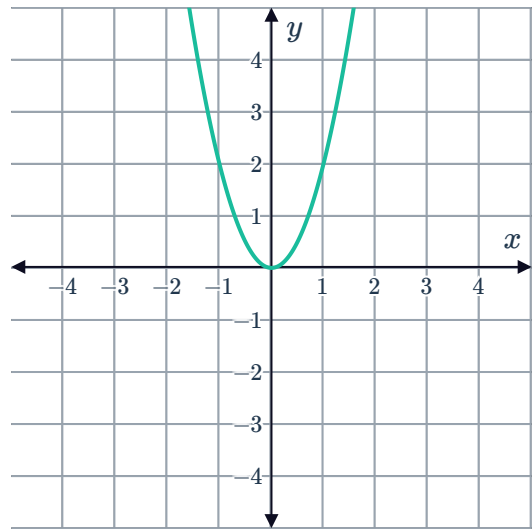
Solve for the constant of proportionality k . Round your answer to the nearest integer.

- 5 y is a power function which is directly proportional to x^n .
 - a Using k as the proportionality constant, form an equation for y in terms of x .
 - b The table below shows values that satisfy the relationship $y = kx^n$, for the case when $n = 3$.

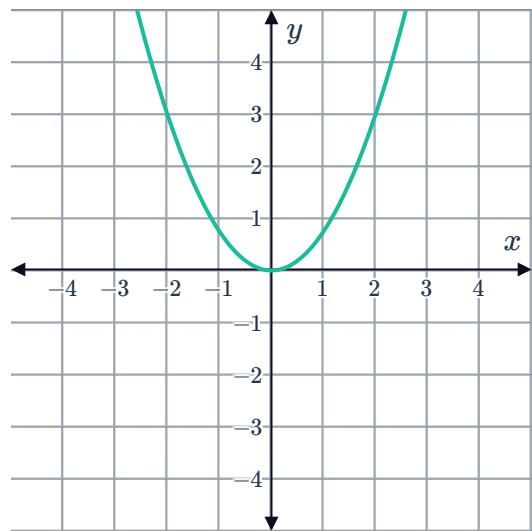
x	2	4	6	8	10
y	-48	-384	-1296	-3072	-6000

Solve for the constant of proportionality k . Round your answer to the nearest integer.

- 6 For the quadratic function pictured here, determine the constant of proportionality k :

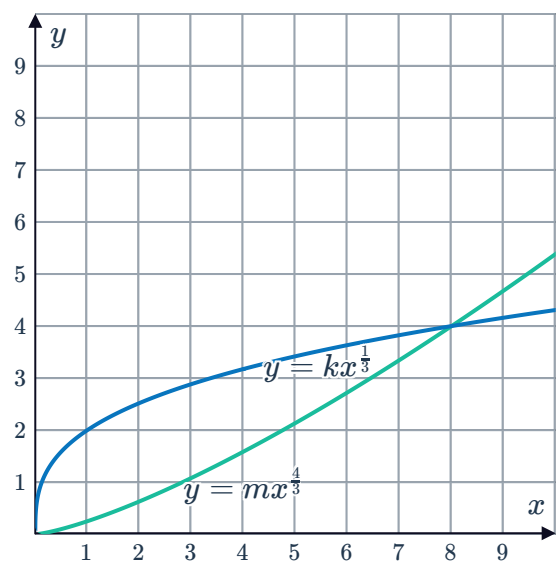


- 7 For the quadratic shown, determine the constant of proportionality k :



- 8 The following graph shows the intersection of the two functions $y = mx^{\frac{4}{3}}$ and $y = kx^{\frac{1}{3}}$:

- Find the value of m .
- Find the value of k .





Applications

- 9 In exchange rate calculations the constant of proportionality is the exchange rate itself. Determine the exchange rate, k , if someone can purchase:
- a 889.00 AUD with 700.00 USD.
 - b 553.00 USD with 700.00 AUD.
- 10 The heart mass H of a mammal is proportional to its body mass M .
- a Using k as the constant of proportionality, form an equation relating H and M .
 - b Given that a human with a body mass of 77kg has a heart mass of 0.462kg, determine k , the constant of proportionality.
 - c If a Chimpanzee has a body mass of 57kg, what is the heart mass H ?
 - d If a Macaque has a heart mass of 0.084kg, solve for its body mass M .
- 11 The blood circulation time C of a mammal is proportional to the fourth root of its body mass B .
- a Using k as the constant of proportionality, form an equation relating C and B .
 - b Given that an elephant with a body mass of 5290 kg has a blood circulation time of 148 seconds, determine k , the constant of proportionality. Round your answer to one decimal place.
 - c If a Orangutang has a body mass of 40 kg, find C , its circulation time. Round your answer to one decimal place.
 - d If a Squirrel has a circulation time of 22 seconds, what is the body mass? Round your answer to one decimal place.